

Leveling the Playing field in the Social Sciences

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The hybridization of digital technologies with society is a novel development within civilization. In this age, the level of interconnection offered by a prosthesis of electronic multimedia communication devices with humans is ushering an unprecedented wave of evolutions. These evolutions appear to take place at rapid timescales involving a widespread and non-geographically localized network of human and non-human actors. Existence is evermore characterized by a participation in digital ecosystems especially with regard to socialization and survival. These observations and areas of inquiry are within the domain of social sciences; Yet, the pace and complexity of current socio-technological evolutions calls for the development of new research technologies and techniques. This paper discusses the use of agent-based testing methods on algorithmic selection based media platforms with intent to explore its theory, design principles, and challenges.

What is Agent-Based Testing?

Agent-Based Testing includes the development of computer-agents or automated scripts running in isolated environments designed to “systematically var[y] the way web pages are accessed... [to] emulate entire human browser sessions ... [and] surf behavior.”(Haim, p. 902). This strategic simulation of “micro-level decision-making within algorithmically curated media environments”(Haim, p. 901) is being used to “describe and articulate the role of algorithms in communication systems and their reactions to users, such as in media distribution, targeted advertising, content recommender systems, and search engines”(Diakopoulos et. al, 2023, p. 2)

This methodology originated in “automated testing procedures for quality assurance within software development, where software reactions toward various kinds of (malformed) input are provoked, logged, and evaluated” (Haim, 2020, p. 901). Adapting this towards computational social audits can empirically address “how algorithmic decisions may be biased or otherwise lead to errors and harms to individuals or society for which they should be held accountable.”(Diakopoulos et. al, 2023, p. 2) The increasing reliance on dynamic, artificially-learned designs in widespread software services encourages “the use of methods like reverse engineering to shed light on the detailed interplay of input, throughput and output.” (Just & Latzer, 2016, p. 16)

Applying Actor-Network Theory in Empirical Media Studies Research

In developing experiments and evaluating results for agent-based tests, researchers should seek out and make transparent the social models informing their designs. Actor-Network Theory (ANT), an approach to social theory that originated in the domain of science and technology studies (STS) in the 1980s, “conceptualizes actors as constantly transforming socio-technical points of intersections in networks and as relations and hybridizations of both technological and social elements.”(Spöhrer, 2016, p. 5) The increased utility of smart-devices as human digital appendages has altered the agency of non-human actors in the definition of social order. Arguably, social order can be better described through the “new and constantly changing forms of digitized cyborg assemblages”(Lupton, 2015, p. 174) where “structures do not simply reside in the actions of people,... [but] in a network of heterogeneous material arrangements.”(Law, 1990, p.16) In reference to ANT’s definition of the social, “applications based on algorithmic selection are characterized by the coevolution of humans (developers, user)

and non-humans (algorithms) within a networked information society... where humans shape algorithms and are simultaneously shaped by them.”(Just & Latzer, 2016, p. 18) Consequently, agent-based test designers are purposely “attentive to the ways that the institutional actors try to mould and shape that behaviour,... structure the media network, and how this allows for certain interactions while blocking others (Teurlings, 2013, p. 11) From this light, “the strength of an application of ANT in Media Studies is its potential to scrutinize “media processes”... [i]nstead of treating media and the process of mediatization as “black boxes,” those processes and the relations to the elements that condition and produce those processes can be “unblackboxed,”.”(Spöhrer, 2016, p. 8-11) The technological tools used in agent-based testing are valuable to a sociology tasked with “characteris[ing] these networks in their heterogeneity, and explor[ing] how it is that they come to be patterned to generate effects like organisations, inequality and power.’ (Law, 1992, p. 381)

Humanness in Artificial Agents

ANT’s recognition that human and non-human agents exist within self-forming and dynamic networks is a useful model for thinking about the organization of users, media objects, and algorithmic curation structures present within social media systems. Internet user’s “Bodies and identities are fragmented into a series of discrete components”(Lupton, 2015, p. 35) enabling softwares to form sophisticated models from an entire history of time-stamped interaction data comprising a digital fingerprint of that human actor. With such capacity to record each digital exchange initiated between the user and platform as well as a plethora of possible user actions, these digital footprints are quite unique, and “not only underpin but actively constitute selfhood, embodiment, social life, social relations and social institutions.”(Lupton, 2015, p. 174) Consequently, all agent-based tests will include a mechanism to achieve degrees of

representativeness for the human actors and digital situations it intends to evaluate. A fundamental challenge for agent-based test designers is to avoid simulating “behaviors that are not plausible with respect to actual human communication behaviors” since doing so makes it “...unclear whether the results would be meaningful in the context of regular human interaction with the machine communicator.” (Diakopoulos et. al, 2023, p. 12) Further, “the fidelity of the simulated agent includes not only its communicative behavior but also its sensed context, such as what location information it makes available to the machine communicator (Haim, 2020). Thus, Agent-based test researchers should make transparent the systematic approach employed towards simulating representative scenarios including aspects both in an agent’s interaction behavior and the characteristics of its implicit metadata.

Advantages with Automation

As opposed to “case studies, expert interviews and discourse analysis,... agent-based solutions enable quantitative research on a broader scale... allow[ing] for a quick and reliable assessment of actual algorithmic reactions to (presumably) human behavior. (Haim, 2020, p. 902) This is achieved by a research methodology centered around the capturing of digital artifacts from the web, databases, and Application Programming Interfaces (APIs), while simulating human-like interaction patterns guided by intelligent multimedia processing tools. One example is an agent-based test designed to assess the “quality, source diversity, and comprehensiveness of news information conveyed by smart speakers”(Diakopoulos et. al, 2023, p. 4) such as Amazon Alexa and Google Home. Relevant news-related user query topics were generated from the “top 20 U.S. Google daily trending search topics collected each day” and formed into query phrases based on crowdsourced research on “four common phrases that people use when seeking news-related information from voice assistants.” Amazon’s language model

Polly was used to “synthesize the text of the user queries to speech... [and] used Amazon Transcribe to transcribe the audio responses and recorded both responses and queries in a database for analysis.”(Diakopoulos et. al, 2023, p. 4 - 5). This automated pipeline, consisting of several stages of data-acquisition, web-interaction, and data-transformation serviced the researchers in executing a highly controlled experiment generating 2,880 query responses over a period of 2 weeks. A final stage of manual human discussion was needed to evaluate the quality of responses generated by the smart devices to these news queries; however, this is a standard approach in grading the informational quality dimensions of news sources such as relevance, accuracy, and timeliness. While not all stages of an agent-based testing pipeline should be automated, many can be which significantly reduces research efforts and ensures a level of control and scale not possible through other methods.

Methodological Challenges

With agent-based tests, “devising the scope of an audit study is a dance between the research questions of the auditor, the ability to collect data about the system that is responsive to those questions, the design and engineering bounds of the system, and the level of resources that can be put towards the audit.” (Diakopoulos et. al, 2023, p. 10) Any deployed system of agents exercising and monitoring a media platform will face limits in its ability to simultaneously perform representative, reactive interactions while collecting and processing metadata on those interactions. For example, an agent-based test designed to characterize the exposure of news content to users on Tik-Tok involved agents simulating varying degrees of news interest on a scale of 0-1. This numeric probability of news interest was used to determine if the agent should watch or skip each video recommended to it by the Tik Tok’s For you page. However, the AI based video processing tools used to assess a video’s news relevance “takes up to 30 seconds,

depending on the length of the video... [and] this runs the risk that TikTok's algorithm picks up an erroneous signal of interest in longer videos." (Diakopoulos & Hagar, 2023, p. 18) Thus, processing limitations in a system, particularly the AI tools which extract meaningful content-specific qualities of a media object in question and inform an agent's generative actions, will force trade offs in an experiment's scope. In other words, this processing limitation potentially excludes representation for users who typically scroll through Tik Tok at faster timescales than the system's comprehensibility. Yet, if desired, agents can be designed to make interaction decisions using metadata with less computationally expensive methods. This may allow for representation of quick scrolling, at the cost of lower resolution or granularity of data made available to the agent at time of interaction-decision.

Intrinsic Agent Metadata

Not only does agent-based testing depend on triggering human-like interactions with a media system, but it also requires intentionality in decorating an automated agent's 'appearance' with respect to the algorithmic system it is auditing. When interacting with any web-service, a set of metadata is exchanged between the user and server which alters the platform's perception of that user. Agent-based test designers ensure that HTTP demographic qualities such as "the language, character-set encoding, time-zone settings, and requested content type to be set accordingly (i.e., through the HTTP-header fields "Accept-Language", "Date", "Accept-Charset", and "Accept")" (Haim, 2020, p. 905) to meet the scope of their design objectives. Additionally, software identity information and internet history are intentionally modified because "information on browsers and operating systems as well as previously set cookies allow for the identification of specific devices and users, even without an actual login" (Haim, 2020, p. 905) Consequently, research groups would benefit from developing scalable and

programmable cloud computing infrastructures with “well-maintained virtual tunnels (VPN).. to increase external validity by mimicking actual user locations (Haim, 2020, p. 903 - 904) A research group investigating multiple agent-based test initiatives should develop effective mechanisms in disseminating jobs among its shared computing resources. These jobs should exercise “a website’s full functionality and also not raise the server’s attention” while including “emulation which also allows for JavaScript execution.” (Haim, 2020, p. 905) Detailed structuring of the internet traffic metadata shared between agents and algorithmic media systems will require high costs in infrastructure development, but will be crucial toward customizable demographic representation.

API TOS Legality

Widespread media platforms such as Meta, X, and Spotify provide a set of tools (APIs) supporting the community of developers who wish to interface with their services and are often used in research projects. These APIs offer formalized mechanisms of access to features of a media platform and its intrinsic data. Unfortunately, these APIs are blackboxed and endowed with functionalities solely permitted by the platform. To this end, API designers’ hold all the power in how a developer or researcher can obtain knowledge from their platform. This may be for infrastructural and resource preservation reasons such as not overloading query requests to servers; however, the types of data available through APIs and limitations in the frequency of access does not always meet the needs for customized research experiments. Further, there is a level of uncertainty in developing research infrastructures dependent upon official APIs. First, companies can change their API features and format at any time which can break substantial functions of code relied upon in stages of a research pipeline. Second, the data made available through APIs is always a subset, and thus a catered sample of the entirety of data existing within

the platform's databases. For these reasons, web scraping is often a more dependable tool in acquiring meaningful data from a media platform; however, it is also not free from risks. The development of automated systems to scrape the web are often explicitly banned from a platform's terms of service and if recognized, can result in account or IP address bans halting research. However, "carefully crafted software" designed with a knowledge of bot detection mechanisms and rate limits "can circumvent such restrictions"(Computational Research in the Post-API Age, DEEN FREELON, 4). Further, the legal implications of performing research in violation of a company's terms of service have been a historically gray area due to the Computer Fraud and Abuse Act of 1986 (CFAA). Yet in the case of *Sandvig v. Barr*, the US court for the District of Columbia ruled that "the CFAA does not criminalize mere terms-of-service violations on consumer websites and, thus, that plaintiffs' proposed research plans" involving the use of autonomous web scraping to assess race and gender discrimination in employment sites "are not criminal under the CFAA." (*Sandvig v. Barr*) Thus, civil disobedience of software Terms of Service to design artificial data scraping systems may be the best approach when used for ethical research.

Conclusion

This paper has explored the fairly new research methodology of agent-based testing with regard to its theoretical underpinnings, technical process, and challenges. As technology has advanced, "Google, Facebook and Amazon as well as many other companies and agencies have become expert at managing data collection, archiving and interpretation in ways about which sociologists and other social scientists working in higher education can only dream." (Lupton, 2015, p. 5) Academic attention is needed to address the level of power large corporations have

in designing artificially intelligent systems which shape platform user's individual development. Traditional efforts towards algorithmic governance may not be successful as "even programmers and software engineers increasingly do not know what 'their' algorithmic selection produces. Hence the discussed disclosure of source codes might often not be a sufficient solution for more democratic control, and there are hardly any other public measures to control the morality in algorithmic reality construction." (Just & Latzer, 2016, p. 19) Agent-based tests have potential to empirically evaluate highly impactful, yet elusive systems in ways that existing methods are not equipped to handle. In the words of Deleuze, "It is not a case of worrying or hoping for the best, but of finding new weapons." (Deleuze, 1992, p. 4)

References

- Brooker, P. D. (2019). *Programming with Python for Social Scientists*. Sage.
- Deleuze, G. (1992). Postscript on the Societies of Control. *October*, 59, 3–7.
<http://www.jstor.org/stable/778828>
- Diakopoulos, N., Bandy, J., & Dambanemuya, H. (2023). Auditing Human-Machine Communication Systems Using Simulated Humans. In *Handbook of Human-Machine Communication* (pp. 178–185). essay, Sage Publications Ltd.
- Freelon, D. (2018). Computational Research in the Post-API Age. *Political Communication*, 35(4), 665–668. <https://doi.org/10.1080/10584609.2018.1477506>
- Hagar, N., & Diakopoulos, N. (2023). Algorithmic indifference: The Dearth of News Recommendations on TikTok. *New Media & Society*, 0(0).
<https://doi.org/10.1177/14614448231192964>
- Haim, M. (2020). Agent-based testing: An Automated Approach toward Artificial Reactions to Human Behavior. *Journalism Studies*, 21(7), 895–911.
<https://doi.org/10.1080/1461670x.2019.1702892>
- Just, N., & Latzer, M. (2016). Governance by Algorithms: Reality Construction by Algorithmic Selection on the Internet. *Media, Culture & Society*, 39(2), 238–258.
<https://doi.org/10.1177/0163443716643157>
- Law, J. (1990). Introduction: Monsters, Machines and Sociotechnical Relations. *The Sociological Review*, 38(S1), 1–23. <https://doi.org/10.1111/j.1467-954x.1990.tb03346.x>
- Law, J. (1992). Notes on the Theory of the Actor-Network: Ordering, Strategy, and Heterogeneity. *Systems Practice*, 5(4), 379–393. <https://doi.org/10.1007/bf01059830>

Lupton, D. (2015). *Digital Sociology*. Routledge.

Sandvig v. Barr, Sandvig v. Barr — Memorandum Opinion (UNITED STATES DISTRICT COURT FOR THE DISTRICT OF COLUMBIA March 28, 2020). Retrieved April 23, 2024, from <https://www.aclu.org/documents/sandvig-v-barr-memorandum-opinion>.

Spöhrer, M. (2017). Applications of Actor-Network Theory in Media Studies: A Research Overview. In *Applying the Actor-Network Theory in Media Studies* (pp. 1–19). essay, IGI Global.

Teurlings, J. (2013). Unblackboxing Production: What Media Studies Can Learn from Actor-Network Theory. In *After the Break: Television Theory Today* (pp. 101–116). essay, Amsterdam University Press.

Woolgar, S. (1990). Configuring the User: The Case of Usability Trials. *The Sociological Review*, 38(S1), 57–99. <https://doi.org/10.1111/j.1467-954x.1990.tb03348.x>